



New City College
Teaching & Learning Lab



Evaluating the Impact of AI-Driven Large Language Models on Teaching Practices in Further Education (UK)

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Abstract:

This study explores the role of AI tools, particularly those powered by Large Language Models (LLMs), in transforming teaching within Further Education (FE) in the UK. Focusing on TeacherMatic, an AI-driven planning tool, it evaluates real-world usage by lecturers through a mixed-method approach at New City College. The research analyses engagement trends, shifts in staff perception, and practical outcomes of AI adoption. Key findings reveal that AI can support efficient, personalised teaching while raising important ethical considerations around data use. The study offers practical recommendations for integrating AI tools into FE teaching in a responsible and impactful way.



This research explores how AI-driven Large Language Models (LLMs), particularly TeacherMatic, influence teaching practices in UK Further Education (FE).

Introduction:

Problem: While AI is widely studied in higher education, its application in FE remains under-researched, especially real-world classroom use.

Research question: Can LLMs effectively enhance teaching and learning in FE? Focus and

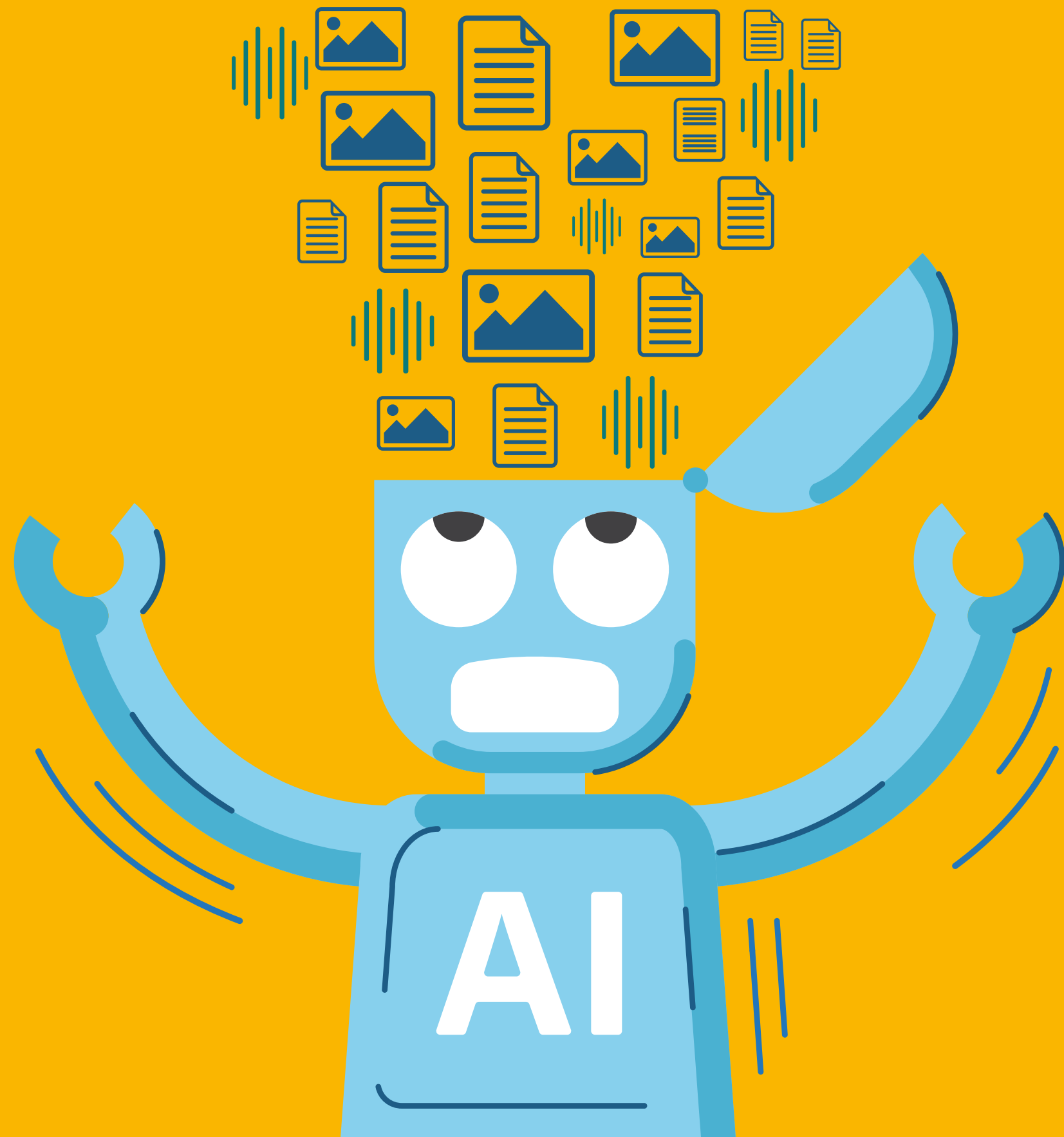
Boundaries: The study focuses on one FE institution, using surveys and usage data. It does not explore technical AI mechanisms.

Fit with other research: Builds on AI-in-education literature, offering new empirical insights specific to the FE context.

Objectives:

1. Investigate reasons for the rise or decline in TeacherMatic users.
2. Evaluate the perception of AI tools (TeacherMatic, ChatGPT, and Copilot) among educators.
3. Identify strategies to enhance the adoption and sustained use of AI tools on FE.

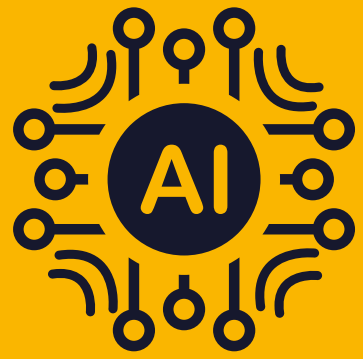
How AI is Revolutionizing Education



Narrow AI is designed to handle specific tasks. It includes marking assignments, creating lesson plans, or supporting teachers with routine work. Tools like TeacherMatic are good examples of this type as these are designed for education-focused tasks (TeacherMatic, 2025).

General AI refers to a broader kind of system that can perform human-like intellectual tasks. However, this type is still theoretical and has not been achieved yet (Holmes et al., 2019).

Super AI is a future concept where machines could potentially surpass human intelligence. Like Gen AI, this idea exists mostly in speculation and theory at this point (Holmes et al., 2019).



LLMs in Education



The UK Department of Education (2024) has published a recent report which confirms that many teachers are starting to use generative tools to personalise learning and save time on repetitive tasks. This means more time and energy for teaching. LLMs such as TeacherMatic help with lesson plans and routine tasks allowing teachers more time to focus on students in live classrooms. Research has shown that these tools can make a difference but how and to what extent remains unexplorable.

For example, one study showed that students were 30% more engaged in tutoring sessions led by LLMs. Students found LLMs support better for tricky STEM topics (Chen et al., 2023). This means that LLM could be a great support in Further Education (FE), where students learn at different speeds and come from varied backgrounds (El Fathi et al., 2025).

LLMs



TeacherMatic: An Ultimate Tool for Educators

TeacherMatic is a digital platform designed to support teachers and other staff in colleges. It helps with a range of tasks e.g. planning, teaching and assessment activities, etc. Since it was launched in April 2023, it has been adopted by over 60% of Further Education colleges. According to the Association of Colleges (2024), teachers save 3.8 hours of work per week on average.

Methodology & Ethical Consideration

Research Design: A mixed-method approach was used, combining quantitative data (usage logs) and qualitative data (surveys) to evaluate AI tool adoption in Further Education.

Type of Study: Mixed-method: integrating both qualitative insights and quantitative analysis.

Population & Sampling: Targeted 601 teaching staff at New City College across four campuses. 73 responses were collected via stratified email distribution.

Data Collection & Analysis: Surveys, institutional reports, and TeacherMatic login data were analysed using statistical tools (Chi-square, ANOVA) and thematic coding (Python in Google Colab).

Ethical Statement: Ethical approval was obtained. Participation was voluntary, anonymous, and GDPR-compliant, ensuring no harm to participants.



Data Collection Methods

Data was collected from previously conducted surveys and pilot interventions on AI at NCC from 2024-2025.

Previous Surveys such as “TeacherMatic Review Survey”, “AI Mid-Pilot Survey”, and “End of AI Project Survey” are used, and secondary data was collected by:

- Review of existing research papers, institutional internal surveys, reports, and AI tool user manuals to contextualise survey findings.
- TeacherMatic Usage Data from Jan-Feb 2025: User login data for 162 active users at New City College was obtained on three separate dates: 07 Jan 2025, 23 Jan 2025, and 13 Feb 2025. This data helps analyse trends, engagement, and adoption patterns over time.

New Survey compares and evaluates AI tools, covering:

- Frequency and ease of use
- Perceived benefits and challenges
- Impact on teaching and learning outcomes
- Ethical considerations (data privacy, bias)



ProS					
Site	Agency	Lecturer	Manager	Curriculum	Grand Total
Ardleigh Green	63	86	10	15	174
Bsix	21	57	7	2	87
Epping Forest	43	55	4	15	117
Hackney	69	124	46	29	268
Havering Sixth Form	17	115	3	15	150
Poplar	59	118	25	27	229
Rainham Construction	16	32	1	7	56
Redbridge College	56	81	8	21	166
Grand Total	344	668	104	131	1247

Table 1: Target Population - Staff Data obtained from New City College's Management Information System (MIS)

Target Audience is 'teaching staff':

- Current and former TeacherMatic users
 - Lecturers, Curriculum Managers (CM), and Senior Leadership Team (SLT)
 - Lecturers not using AI tools but are open to adopting them
- 601 educators were targeted from Havering Sixth Form, Poplar, Rainham Construction, and Redbridge campuses. A total of 73 responses were collected over 23-days, resulting in a response rate of 12.15%.



Evaluating the Impact of AI-driven Large Language Models on Teaching Practices in FE

"This survey aims to explore the role of AI tools such as **TeacherMatic**, **ChatGPT**, and **Copilot** in teaching practices within UK Further Education (FE). Your responses will help assess the effectiveness, challenges, and future potential of AI-powered tools in lesson planning, resource creation, and student engagement. The survey should take approximately **10-15 minutes** to complete. Your responses will remain anonymous, **confidential** and will be used solely for research purposes."

Thank You for your valuable contribution!

By Saadia Adnan

Python Data Analysis

```
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# Install required libraries (if not already installed)
!pip install pandas scipy matplotlib seaborn

# Import necessary libraries
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from scipy.stats import chi2_contingency

# Sample data: Cross-tabulation of AI Tools vs Frequency of Use
# These values are estimated based on the survey images provided
data = {
    'Daily':      [4, 1, 1, 0, 0], # ChatGPT, TeacherHolic, Copilot, Other, None
    'Weekly':     [14, 6, 4, 2, 0],
    'Monthly':    [8, 5, 3, 2, 1],
    'Rarely':     [10, 3, 0, 1, 0],
    'Never':      [2, 1, 0, 1, 0]
}

# Row Labels
tools = ['ChatGPT', 'TeacherHolic', 'Copilot', 'Other', 'None']

# Create DataFrame
df = pd.DataFrame(data, index=tools)

# Display data
print("Contingency Table (AI Tool vs Frequency of Use):")
display(df)

# Chi-Square Test of Independence
chi2, p, dof, expected = chi2_contingency(df)

# Output test results
print(f"Chi-Square Statistic: {chi2:.2f}")
print(f"Degrees of Freedom: {dof}")
print(f"P-value: {p:.4f}")
```

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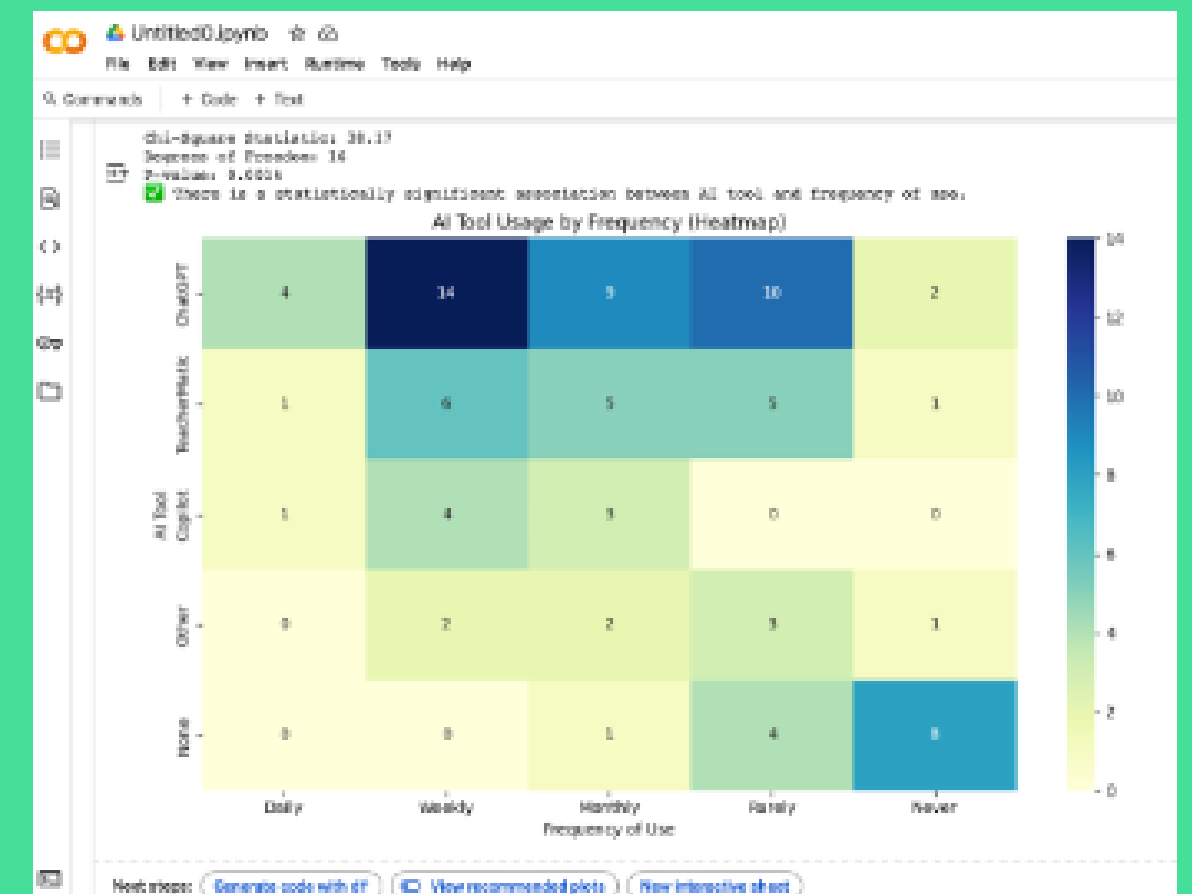
# Print P-value
print(f"P-value: {p:.4f}")

if p < 0.05:
    print("There is a statistically significant association between AI tool and frequency of use.")
else:
    print("There is no statistically significant association between AI tool and frequency of use.")

# Visualize with a heatmap
plt.figure(figsize=(10, 8))
sns.heatmap(df, annot=True, cmap='magma', cbar=True)
plt.title("AI Tool Usage by Frequency Heatmap")
plt.xlabel("Frequency of Use")
plt.ylabel("AI Tool")
plt.show()

# Contingency Table (AI Tool vs Frequency of Use)
df

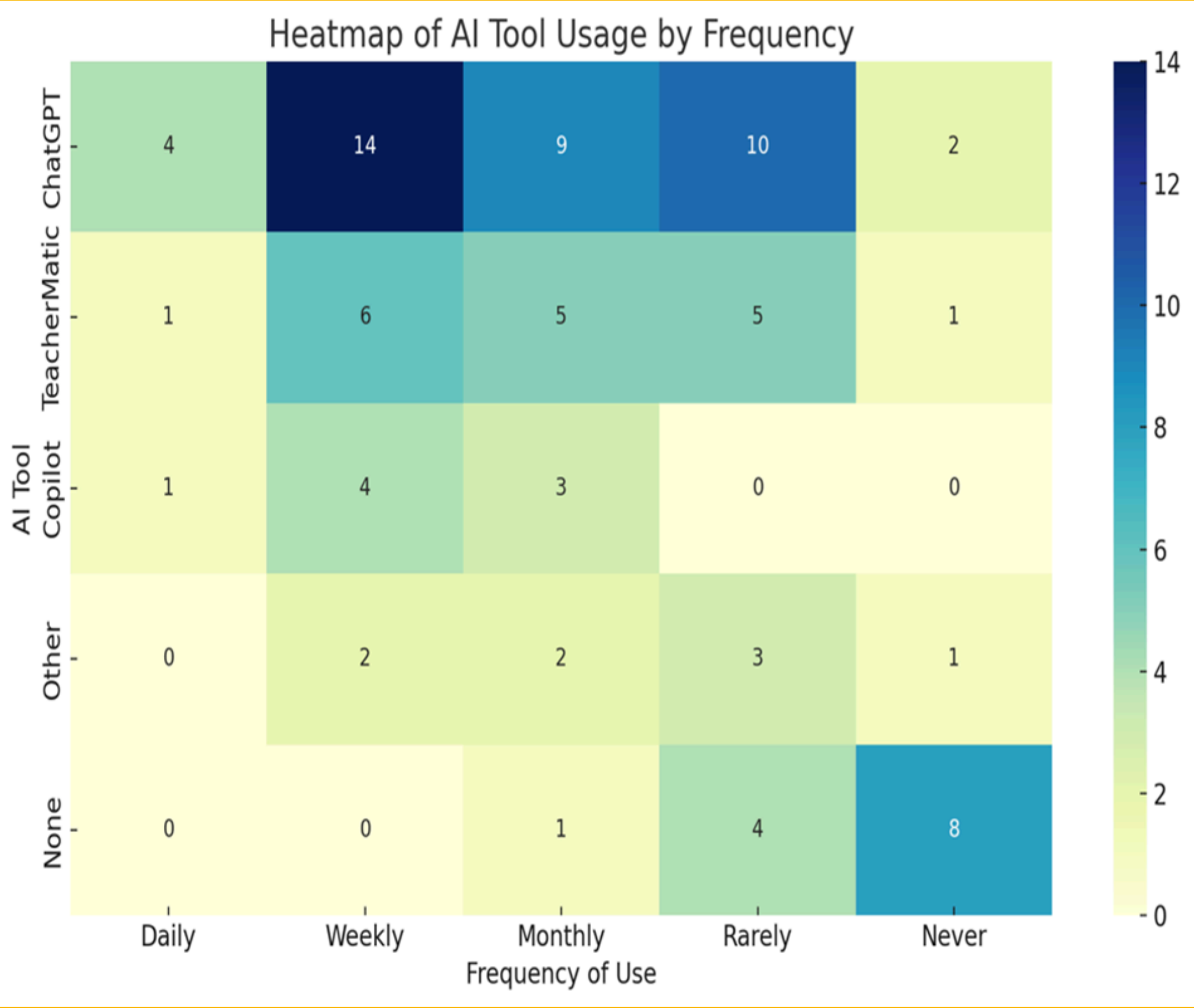
# Chi-Square Statistic: 38.17
# Degrees of Freedom: 14
# P-value: 0.0000
```



Data analysis Results

Statistical Testing

To examine the association between categorical variable, a Chi-square test of independence is conducted. For comparing group means, a one-way ANOVA is used (Agresti & Kateri; 2022).



Chi-Square Test Heatmap of AI Usage by Frequency

Chi-Square Test Result

Chi-square statistics = 38.17
Degree of freedom (df) = 16
p-value = 0,0014

SAMR Model

SAMR Level	Example from Study	Commonality
Substitution	Using TeacherMatic to auto-generate lesson plans	Very common
Augmentation	LLM-generated differentiated resources for mixed-ability learners	Common
Modification	ChatGPT used for simulating dialogue or peer-based roleplay tasks	Occasional
Redefinition	LLM supported reflective journaling or curriculum mapping with learners	Rare

Discussion/Conclusion (Strengths and Limitations)

- **Findings & Hypothesis:** The study supports the hypothesis that AI-driven Large Language Models (LLMs), particularly TeacherMatic, enhance teaching practices in Further Education (FE). Educators reported improved lesson planning, reduced administrative workload, and moderate gains in learner engagement. Statistical analysis confirmed a significant relationship between tool type and usage frequency.
- **Comparison with Literature:** These findings align with prior research (e.g., Gan et al., 2023; Holmes et al., 2019), which highlights AI's potential to streamline educational tasks. However, unlike higher education, FE adoption remains cautious, with most usage limited to planning and resource creation.
- **Significance:** This study contributes original empirical evidence from the under-researched FE sector. It offers practical insights into AI integration, informing institutional strategies, professional development, and ethical considerations for AI use in education.
- **Limitations:** The study is limited to one institution (New City College), which may affect generalisability. The 12.15% survey response rate introduces potential bias, and reliance on self-reported data may not fully reflect actual usage. Additionally, the absence of student perspectives and the short evaluation period limit the depth of impact assessment.

Future Recommendations

- Conduct multi-institutional studies to improve generalisability across the FE sector.
- Implement longitudinal research to assess long-term impacts of LLMs on teaching and learning.
- Include student perspectives to evaluate learner engagement and outcomes.
- Explore subject-specific effectiveness of AI tools across STEM, humanities, and vocational areas.
- Develop ethical frameworks and data privacy protocols tailored to AI in education.
- Investigate effective CPD models to enhance AI literacy among educators.
- Compare AI tools (e.g., ChatGPT, Copilot, TeacherMatic) to assess usability, effectiveness, and educator preferences.

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Thank You!

